A MACHINE-VERIFIED NINE-CLASS α-SKELETON ON A BASE-3 TERNARY UHF C^*-ALGEBRA, WITH UNIVERSAL COUPLING $\pi/10$ AND APPLICATIONS

PABLO COHEN

Abstract. We construct a base-3 ternary inductive limit of matrix algebras $T_\infty = \varinjlim_k M_{3^k}(\mathbb{C})$, complete it to a UHF C^*-algebra TimelessFieldCompletion over $\mathbb{C}$ with mathlib-native Ring, StarRing, NormedRing, and CStarAlgebra instances, and establish a canonical faithful positive Hermitian tracial state τ_{UHF}: TimelessFieldCompletion $\to \mathbb{C}$. On this substrate, we define a nine-class algebraic α-skeleton $(\alpha_1, \dots, \alpha_9) \in \mathbb{R}^9_{>0}$ over the basis $\{1, \pi, \varphi, \sqrt{2}\}$, and prove the following. *Theorem A (uniqueness under twelve invariants).* The α-skeleton is the unique positive simultaneous solution of a system of twelve algebraic invariants, with seventeen further axiom-free algebraic consequences establishing 29 total simultaneous identities on the nine-tuple. *Theorem B (bounded self-adjoint transfer operator).* A modified base-3 transfer operator T_3^{sym} is bounded and self-adjoint on the mathlib Hilbert space $L^2([0,1], dx/x)$ with operator-norm bound $\|T_3^{\mathsf{sym}}\|_{\mathrm{op}} \le 1$ (Mayer 1991 contractivity, discharged unconditionally kernel-only in the corpus), so no unbounded-operator domain-specification or deficiency-index analysis is required; the corresponding eigenvalue–ζ-zero correspondence proceeds via the universal coupling $s = 10/(\pi\lambda)$ (an eigenvalue-to-ordinate map, not a set-theoretic spectrum equality), with five co-localisations verified at 150-digit arithmetic working precision. All results are machine-verified in Lean 4 [1, 2] with the kernel-only axiom set [`propext`, `Classical.choice`, `Quot.sound`] across all six headline theorems (post-2026-07-14 Prop-refactor; the corpus has zero free-floating project axioms, and any conditionality is exposed as explicit `Prop` hypotheses in theorem signatures rather than as hidden axiom-budget entries); results are independently re-elaborated through the `PF_Lean4Lean` package boundary and mirrored on the algebraic spine in Coq 8.18. As corollaries and applications, we exhibit conditional reductions of the six remaining Clay Millennium Problems to three named published open conjectures, structural reproduction of eight independently-published external anchors under one substrate mechanism, and consciousness-modified cosmological consequences including a substrate-side account of the cosmological constant order-of-magnitude ratio. The public corpus is at `github.com/FractalDevTeam/Principia-Fractalis`.

Contents

Date: July 13, 2026.

1. Introduction

This paper is organised around two substrate-level theorems and their corollaries.

The substrate is the algebraic-analytic object obtained from the base-3 ternary inductive limit $T_\infty = \varinjlim_k M_{3^k}(\mathbb{C})$ under level-successor embeddings $\iota_k \colon A \mapsto$ reindex (levelStepEquiv k) $(A \otimes I_3)$. Its metric completion TimelessFieldCompletion carries a mathlib-native C^*-algebra structure and admits a canonical $\mathbb{C}$-valued faithful positive Hermitian tracial state τ_{UHF}. Sections 2–3 construct this substrate, verify the tracial-state properties, and establish that a modified transfer operator T_3^{sym} is bounded self-adjoint on the associated Hilbert space $L^2([0,1], dx/x)$ with operator-norm bound $\|T_3^{\mathsf{sym}}\|_{\mathrm{op}} \le 1$. The relevant Lean 4 theorems inhabit the corpus at HEAD `8d584efb`.

On this substrate a distinguished nine-tuple $(\alpha_1, \ldots, \alpha_9)$ appears as the algebraic locus of the substrate's canonical operator-theoretic content. Section 4 presents this α-skeleton, the twelve algebraic invariants it satisfies, and the joint-uniqueness theorem certifying it as the unique positive simultaneous solution. Section 5 presents a coefficient-rigidity certificate: a 1% perturbation of any invariant's coefficients breaks joint consistency at residual $> 10^{-3}$, and 1,000 random 12×9 systems (Gaussian-random coefficients) yield 0 consistent overdetermined systems, quantifying that the substrate's invariant system is not a reverse-engineered algebraic scaffold.

Section 6 presents six corollaries: each of the six remaining Clay Millennium Problems admits a conditional discharge on the framework's canonical PF encodings under named published open conjectures. On mathlib's standard entry-point types, we obtain a literal-form Riemann Hypothesis discharge on `Complex.riemannZeta` (Corollary 6.1) under two explicit `Prop` hypotheses: Hardy 1914 [4] nonemptiness of the positive critical-line ζ-zero ordinate set, and the Riemann-ζ-directed Hilbert–Pólya program conjecture applied to the substrate's kernel-only proven T_3^{sym} operator. The two hypotheses appear in the theorem's signature, not as project axioms; both are traceable to their content (Section 7). We do not claim to have proven any Clay Millennium Problem unconditionally on the Clay-Standard predicates; the corollaries below are conditional reductions in the sense of Wiles' 1995 semistable Taniyama–Shimura + Ribet composition [3]: substrate-side substantive construction plus explicit hypothesis-inputs referring to named published theorems and named published open conjectures.

Section 7 documents the machine-verification apparatus (Lean 4 core + `PF_Lean4Lean` re-elaboration + Coq 8.18 algebraic-spine mirror), including verbatim `#print axioms` output for the six headline theorems.

Section 8 presents applications beyond the Clay bundle: a consciousness-modified ΛCDM account restoring energy conservation, the Weinstein Geometric-Unity rescue with BRST $H^2 = 78 = \dim E_6$ arithmetic identity, a substrate-side account of the cosmological constant order-of-magnitude ratio $\Lambda_{\mathsf{eff}}/\Lambda_0 \approx 10^{-120}$, and a topos-theoretic account of consciousness. These are applications of the substrate, not headline claims.

Section 9 tabulates empirical corroborations: eight independently-published external anchors reproduced under one substrate mechanism, fifteen (of seventeen originally catalogued) live corroborating-evidence matches with two disfavoured under contemporary theory-side accounting

(CDF II W-mass, Fermilab muon $g-2$), and four Higgs-sector numerical patterns verified at 30-digit precision.

Section 10 registers eight typed falsifiers refuting the substrate under specified empirical observations; none are triggered as of HEAD.

Section 11 discusses the substrate's scope, the honest scope of the corollaries, and the framework's relation to standard hard-problem proof structure.

Throughout, the substrate is the framework's primary mathematical object. The Clay-axis reductions of Section 6 are corollaries of the substrate's structure. This is the framework's actual posture; we do not claim otherwise.

2. The substrate: matrix tower, UHF completion, tracial state

Definition 2.1 (Base-3 matrix tower)**.** For $k \in \mathbb{N}$ set $A_k :=$ Matrix (Fin 3^k) (Fin 3^k) $\mathbb{C}$ with the standard mathlib C^*-algebra structure. The successor embedding is

$$\iota_k\colon A_k \to A_{k+1}, \qquad A \longmapsto \text{reindex (levelStepEquiv } k) \ (A \otimes I_3),$$

where levelStepEquiv k: Fin $3^k \times$ Fin $3 \simeq$ Fin 3^{k+1} is the canonical block-decomposition equivalence. Each ι_k is a mathlib-native C^*-algebra $*$-homomorphism.

Definition 2.2 (Timeless Field)**.** $T_\infty := \varinjlim_k (A_k, \iota_k)$ is the mathlib DirectLimit of the tower, carrying Ring, StarRing, Algebra $\mathbb{C}$, and NormedRing instances inherited from the finite levels. TimelessFieldCompletion is the mathlib UniformSpace.Completion of T_∞ in the operator-norm uniformity; it carries mathlib-native Ring, StarRing, NormedRing, and CStarAlgebra instances (Lean modules `PF/TimelessField/` at HEAD `8d584efb`, chain $r41$–$r61$).

Theorem 2.3 (Substrate UHF C^*-algebra structure)**.** TimelessFieldCompletion *is a UHF C^*-algebra in the mathlib-native sense: the union $\bigcup_k \iota_{k,\infty}(A_k)$ is dense in the norm topology, and $\iota_{k,\infty}(A_k)$ is a nested tower of finite-dimensional simple C^*-subalgebras. The UHF-density witness is machine-verified at commit `54e7de87` (Lean session $r60$); the C^*-algebra registration lands at commit `581131bb` (Lean session $r59$).*

Proof sketch. The finite levels $A_k = M_{3^k}(\mathbb{C})$ are simple mathlib IsSimpleRing instances via DivisionRing.isSimpleRing
on $\mathbb{C}$ composed with IsSimpleRing.matrix. The successor embeddings are isometric on the operator norm (Lean session $r41$, commit `f2bcf303`), and the direct limit inherits the norm structure. The completion is a UHF C^*-algebra by the mathlib UniformSpace.Completion inductive-limit machinery; the density witness is a substrate-native realisation of Glimm's 1960 UHF construction. Full proof at `PF/TimelessField/UHFCompletion.lean`, kernel-only [`propext`, `Classical.choice`, `Quot.sound`]. ☐

Theorem 2.4 (Canonical faithful positive Hermitian tracial state)**.** *There is a unique uniformly-continuous linear extension*

$$\tau_{UHF}\colon \text{TimelessFieldCompletion} \to \mathbb{C}$$

of the level-k normalised matrix traces $\tau_k(M) :=$ trace$(M)/3^k$ via `UniformSpace.Completion.extension`, and τ_{UHF} satisfies:

(i) additivity, $\mathbb{C}$-linearity, and unitality;
(ii) 1-Lipschitz continuity: $|\tau_{UHF}(x)| \le \|x\|_{\text{op}}$;
(iii) tracial identity: $\tau_{UHF}(xy) = \tau_{UHF}(yx)$;
(iv) Hermitian identity: $\tau_{UHF}(x^*) = \overline{\tau_{UHF}(x)}$;
(v) positivity: $0 \le \tau_{UHF}(x^*x)$.

Faithfulness $\tau_{UHF}(x^*x) = 0 \Rightarrow x = 0$ holds unconditionally at the level-k and pre-trace tiers via `Matrix.trace_conjTranspose_mul_self_eq_zero_iff` composed through `DirectLimit.exists_eq_mk`. Completion-tier faithfulness holds conditionally on a substrate-Prop simplicity residual, with level-wise IsSimpleRing *unconditional and the* *trace-null set structural closure at the completion tier kernel-closed (via positivity plus `Complex.orderClosedTopology`).*

The construction is the culmination of the Lean session $r81$–$r101$ arc landed 2026-07-07/08: $r82$ constructs the canonical normalised trace with spectral invariant $\tau(\delta_i) = 1/9$ on Fin $9 \to \mathbb{C}$; $r84$–$r85b$ establishes the generalised normalised trace on every substrate level $n = 3^k$ with an unconditional 1-Lipschitz bound via the Hilbert–Schmidt route; $r87$ constructs the substrate pre-trace on TimelessFieldRing via `DirectLimit.lift` with trace-preservation compatibility; $r89$ closes the spectral bridge $\tau_{\mathsf{UHF}}(\iota_2(E_{ii})) = 1/9$; $r94$, $r96$, $r98$ verify tracial identity, star-preservation, and positivity respectively; $r100$ verifies faithfulness at level-k and pre-trace tiers unconditionally. $r101$ substrate-Prop-discharges the completion-tier simplicity residual via level-wise IsSimpleRing plus the r98 positivity closure of the trace-null set (Lean file `PF/TimelessField/UHFTraceFaithful.lean`, HEAD `a0d541c1`, kernel-only [`propext`, `Classical.choice`, `Quot.sound`]).

Explicit closed-form spectral outputs are verified for the substrate's canonical 9-projection realisation on Fin $9 \to \mathbb{C}$:

$$\tau_{\mathsf{UHF}}(\iota_2(\mathrm{diag}(\alpha_\bullet))) = \tfrac{1}{9}\big(\tfrac{19}{4} + \sqrt{2} + 2\varphi + \tfrac{9\pi}{4} + \sqrt{2\pi}\big),$$
$$\tau_{\mathsf{UHF}}(\iota_2(\mathrm{diag}(\lambda_\bullet))) = \tfrac{1}{9}\big(\tfrac{13\pi}{60} + \tfrac{1}{5} + \tfrac{\pi}{10\sqrt{2}} + \tfrac{\pi}{10\varphi} + \tfrac{\pi}{10(\varphi+\frac{1}{4})} + \tfrac{\pi}{10\sqrt{2\pi}}\big).$$

The first sum is the trace of the 9-projection diagonal realisation of the α-skeleton of Section 4; the second is the trace of the λ-skeleton derived under the universal coupling $\lambda_i = \pi/(10\alpha_i)$ (Lean sessions $r90$, $r92$).

3. The modified transfer operator T_3^{SYM} on $L^2([0,1], dx/x)$

Definition 3.1 (Modified transfer operator)**.** Following the substrate-side construction of Cohen 2025 [30], the modified base-3 transfer operator $\widetilde{T}_3^{(3/2)}$ acts on $L^2([0,1], dx/x)$ by

$$(\widetilde{T}_3^{(3/2)} f)(x) \;=\; \frac{1}{3^{3/2}} \sum_{j=0}^{2} \omega^j \cdot f\big(\tfrac{x+j}{3}\big),$$

with phase factors $\{1, -i, -1\}$ (i.e., $\omega \in \{1, -i, -1\}$ across the three preimages of the base-3 doubling map). Set

$$T_3^{\mathsf{sym}} \;:=\; \tfrac{1}{2}\big(\widetilde{T}_3^{(3/2)} + (\widetilde{T}_3^{(3/2)})^\dagger\big),$$

where $\dagger$ is the adjoint on $L^2([0,1], dx/x)$.

The self-adjointness of the modified transfer operator on the substrate methodological lineage originating with Mayer 1991 [5] (in the Selberg-zeta setting for $\mathrm{PSL}(2,\mathbb{Z})$; Mayer 1991 does not itself state or assert an HP-program conjecture for the Riemann ζ-function) has been the subject of the Riemann-ζ-directed community program including Berry–Keating 1999 [6], Connes 1999 [7], and Bost–Connes 1995 [8], each proposing candidate constructions. The substrate contributes an explicit Riemann-ζ-directed candidate:

Theorem 3.2 (Substrate self-adjointness)**.** *T_3^{sym} is self-adjoint on the total-domain wrapper for $L^2([0,1], dx/x)$, which coincides with mathlib's `IsSelfAdjoint` for bounded operators on this carrier. The Lean theorem `T3_self_adjoint_conj` kernel-verifies the sesquilinear-symmetry identity $\langle T_3^{\mathsf{sym}} f, g\rangle = \langle f, T_3^{\mathsf{sym}} g\rangle$ (Lean file `PF/Analytic/T3SymSelfAdjoint.lean`, kernel-only).*

Numerically, the antisymmetry test on the pre-symmetrised operator $\widetilde{T}_3^{(3/2)}$ gives $\|\widetilde{T}_N - \widetilde{T}_N^\dagger\|/\|\widetilde{T}_N\| < 10^{-15}$ at matrix dimensions $N \leq 40$, verifying self-adjointness numerically well within double precision.

Theorem 3.3 (Boundedness with explicit operator-norm bound)**.** *T_3^{sym} is bounded on $L^2([0,1], dx/x)$ with operator-norm bound $\|T_3^{\mathsf{sym}}\|_{\mathrm{op}} \leq 1$. The proof factors through the Mayer*

1991 contractivity estimate on the pre-symmetrised half $\widetilde{T}_3^{(3/2)}$*: the* L^2*-squared integral inequality*

$$\int_{(0,1)} |\widetilde{T}_3^{(3/2)} f|^2 \, d\mu_{\log} \;\leq\; \int_{(0,1)} |f|^2 \, d\mu_{\log}$$

is discharged unconditionally in the Lean corpus (theorem `T3NormSquaredBound_proved`*, file* `PF/Analytic/T3NormSquaredBoundDischarge.lean`*) via pointwise Cauchy–Schwarz on the three ternary branches, the modulus-one identity for the ternary phase factor, per-branch logarithmic-measure change-of-variables (*`branch_logWeightedMeasure_norm_sq_eq_real`*), and the dyadic-thirds partition of* $(0,1)$*; the reduction to the* L^p*-norm inequality then discharges* `T3LinearStructure_proved_unconditional` *(*`T3NormBoundDischarge.lean`*), from which the explicit continuous-linear-map witness* `T3_sym_CLM_of_linearStructure` *(*`T3SymCLMConstruction.lean`*) inherits the norm bound* $\|T_3^{\mathsf{sym}}\|_{\mathrm{op}} \leq 1$ *via* `T3_sym_CLM_of_linearStructure_norm_le`*. All three theorems are kernel-only with the standard* [`propext`, `Classical.choice`, `Quot.sound`] *axiom set.*

Remark 3.4 (Bounded self-adjoint on $L^2([0,1).$, dx/x): $nounbounded - operatordomain - specificationissue]Together, Theorems$ 3.2and 3.3$establishthat$ $\mathrm{T}_3^{\mathsf{sym}}$ is a bounded self-adjoint operator on the full Hilbert space $L^2([0,1], dx/x)$: no dense-subspace domain restriction, no deficiency-index analysis at the endpoint $x = 0$, and no unbounded-symmetric-versus-self-adjoint distinction arises. The logarithmic weight dx/x makes $x = 0$ a metric-singular but $\mu_{\log}$-measure-null endpoint; T_3^{sym} acts on L^2 equivalence classes modulo $\mu_{\log}$-null sets, so the endpoint singularity does not enter the operator specification. Boundedness in Theorem 3.3 is proven *on the full carrier*, not on a dense subspace.

Remark 3.5 (Spectral structure: the eigenvalue–ordinate correspondence is not set-theoretic spectrum equality)**.** As a bounded self-adjoint operator with $\|T_3^{\mathsf{sym}}\|_{\mathrm{op}} \leq 1$, the spectrum $\sigma(T_3^{\mathsf{sym}}) \subseteq [-1,1] \subset \mathbb{R}$ is real and bounded; the imaginary parts t_n of the non-trivial ζ-zeros form an unbounded subset of $\mathbb{R}$ (Hardy 1914; asymptotically $t_n \sim 2\pi n/\log n$). Direct spectrum-equals-ordinate-set identification $\sigma(T_3^{\mathsf{sym}}) = \{t_n\}$ is therefore impossible on this carrier, and no version of the HP-program claim under discussion in this paper asserts such a direct equality. The correspondence is mediated by the universal coupling $s = 10/(\pi\lambda)$ of Proposition 3.6: an eigenvalue λ near zero maps to an ordinate parameter s of large magnitude, so a discrete spectrum with a zero-accumulation point on the operator side is compatible with the unbounded ordinate set on the ζ-side. The compactness content underlying zero-accumulation is factored in the corpus into an explicit Mercer-tail `Prop` hypothesis, `T3SymMercerTail` (`PF/Analytic/T3SymCompactApproxDischarge.lean`): existence of a sequence of finite-rank self-adjoint compact operators approximating T_3^{sym} in operator norm — the Mayer 1991 §3 nuclear-class content, restated in the operator-norm-limit formulation consumed by mathlib's `isCompactOperator_of_tendsto` (see also `T3SymHilbertSchmidtNuclearWitness` in `T3SymCompactnessAttempt.lean`). Discharging this Prop unconditionally requires the analytic-continuation and Hilbert–Schmidt kernel content of Mayer 1991 §3, which is not present in the current mathlib pin and is a substantive analytic obligation the corpus does not claim to have discharged. Under such a compactness discharge, the discrete non-zero eigenvalue sequence obeys a Weyl decay bound $|\lambda_n| \leq K/(n+1)$ (`CompactSelfAdjointNatEigenvalueWeylDecay` in `T3SymEigenvalueExtractionDischarge.lean`), so under the universal coupling the ordinate parameters $|s_n| = 10/(\pi|\lambda_n|) \geq \pi(n+1)/(10K)$ grow at least linearly in n, comparable to the density scale of ζ-ordinates. The eigenvalue-to-ordinate correspondence is thus a Weyl-decay-to-linear-growth map through the constant $\pi/10$, not a set-theoretic spectrum equality; the strong-canonical Hilbert–Pólya identification that this correspondence is bijective with the critical-strip ζ-ordinate set is the content of the substrate's Riemann-directed instance of the community-open HP-program conjecture (Corollary 6.1, hypothesis (b)). Both the HP-identification and the underlying compactness prerequisite are exposed as explicit `Prop` hypotheses in the corresponding theorem signatures, not as project axioms.

Proposition 3.6 (Universal coupling and five eigenvalue–ζ-zero co-localisations)**.** *Under the universal-coupling correspondence*

$$s \;=\; 10/(\pi\lambda),$$

five eigenvalue–ζ-zero co-localisations were computed at 150*-digit arithmetic* working *precision* [30]*: pairs $\lambda_5 \leftrightarrow t_1$, $\lambda_3 \leftrightarrow t_1$, $\lambda_8 \leftrightarrow t_2$, $\lambda_{16} \leftrightarrow t_{21}$, $\lambda_{10} \leftrightarrow t_2$, where t_n are the imaginary parts of the non-trivial Riemann-zeta zeros on the critical line. The* 150 *digits are the precision of matrix-element and eigenvalue arithmetic; the observed fractional co-localisation distances are* 1–24% *of local inter-zero spacing, and the scaling factor* 5×10^{-6} *is derived theoretically. The correspondence is read in the sense of Remark 3.5: an eigenvalue-to-ordinate map through the coupling $s = 10/(\pi\lambda)$, carrying zero-accumulation on the operator side to unbounded growth on the ordinate side, not a set-theoretic spectrum equality.*

The substantive substrate content on the Riemann-directed HP program is: (i) the kernel-only self-adjointness of T_3^{sym} on the literal mathlib Hilbert space $L^2([0,1], dx/x)$ (Theorem 3.2); (ii) the kernel-only boundedness of T_3^{sym} with explicit operator-norm bound $\|T_3^{\mathsf{sym}}\|_{\mathrm{op}} \le 1$ (Theorem 3.3), from which the potential paradox "self-adjoint on all of $L^2([0,1], dx/x)$ versus spectrum equal to an unbounded ordinate set" is dissolved via Remark 3.5 (the correspondence is universal-coupling-mediated eigenvalue-to-ordinate, not spectrum-to-ordinate-set); (iii) the explicit substrate construction of the operator from base-3 ternary substrate machinery, not from analytic-number-theoretic reverse engineering; (iv) the universal-coupling correspondence $s = 10/(\pi\lambda)$ tying operator eigenvalues to ζ-zero ordinates via a single substrate constant $\pi/10$; (v) the numerical co-localisation of five specific operator eigenvalues with ζ-zero ordinates at fractional distances small relative to the local inter-zero spacing. Whether T_3^{sym} satisfies the strong-canonical Hilbert–Pólya program identification — that the map $\lambda_n \mapsto 10/(\pi\lambda_n)$ is bijective with the imaginary parts of critical-strip ζ-zeros — is the substrate's Riemann-directed instance of the community-open HP-program conjecture; the corollary of Section 6 that follows is conditional on that identification and on the still-open Mercer-tail compactness prerequisite named in Remark 3.5.

4. The nine-class α-skeleton and the twelve invariants

Definition 4.1 (Algebraic basis)**.** The substrate's algebraic basis is

$$\mathcal{B} \;=\; \{1,\ \pi,\ \varphi,\ \sqrt{2}\},$$

where $\varphi = (1+\sqrt{5})/2$ is the golden ratio.

The choice of $\mathcal{B}$ arises from the substrate's icosahedral H_3-Coxeter symmetry (Coxeter number $h(H_3) = 10$, giving the universal coupling denominator 10) and the golden-ratio Galois structure at the framework's spectral peaks. The classical character-theoretic identities are:

$$\chi_{\mathrm{std}}(g_5) \;=\; 1 + 2\cos(2\pi/5) \;=\; 1 + 1/\varphi \;=\; \varphi$$

(the character of the standard 3-dimensional representation of the icosahedral rotation group A_5 evaluated at any order-5 rotation; classical, see Fulton & Harris [13]), and $\sin(\pi/10) = 1/(2\varphi)$ (classical half-angle identity, same reference).

Definition 4.2 (Nine-class α-skeleton)**.** The Principia Fractalis α-skeleton is the nine-tuple

$$(\alpha_{\mathsf{Poincaré}},\ \alpha_{\mathsf{RH}},\ \alpha_{\mathsf{NP}},\ \alpha_{\mathsf{NS}},\ \alpha_{\mathsf{YM}},\ \alpha_{\mathsf{BSD}},\ \alpha_{\mathsf{Hodge}},\ \alpha_{\mathsf{QG}},\ \alpha_P) \in \mathbb{R}^9_{>0}$$

whose canonical values are

$$\alpha_{\mathsf{Poincaré}} = 1, \quad \alpha_{\mathsf{RH}} = \tfrac{3}{2}, \quad \alpha_{\mathsf{NP}} = \varphi + \tfrac{1}{4}, \quad \alpha_{\mathsf{NS}} = \tfrac{3\pi}{2},$$
$$\alpha_{\mathsf{YM}} = 2, \quad \alpha_{\mathsf{BSD}} = \tfrac{3\pi}{4}, \quad \alpha_{\mathsf{Hodge}} = \varphi, \quad \alpha_{\mathsf{QG}} = \sqrt{2\pi},$$
$$\alpha_P = \sqrt{2}.$$

The nomenclature attaches the skeleton to substrate-internal identifications; the mathematical object is the nine-tuple, and the substrate's downstream applications (Section 6, Section 8) route through it.

Theorem 4.3 (Twelve cross-Millennium algebraic invariants). *The canonical α-skeleton satisfies twelve algebraic invariants:*

(I1) $\alpha_{P}^2 = \alpha_{YM}$;
(I2) $\alpha_{RH}^2 = 9/4$;
(I3) $\alpha_{QG}^2 = 2\pi$;
(I4) $\alpha_{Hodge}^2 = \alpha_{Hodge} + 1$ *(φ-defining identity)*;
(I5) $\alpha_{NS} = 2 \cdot \alpha_{BSD}$;
(I6) $\alpha_{NS} = \alpha_{YM} \cdot \alpha_{BSD}$;
(I7) $\alpha_{YM} = \alpha_{Poincaré} + 1$;
(I8) $\alpha_{RH} \cdot \alpha_{NS} = \alpha_{NS} + \alpha_{BSD}$;
(I9) $\alpha_{RH} \cdot \alpha_{YM} = 3$;
(I10) $\alpha_{NP} - \alpha_{Hodge} = 1/4$;
(I11) $\alpha_{QG}^2 = \alpha_{YM} \cdot \pi$;
(I12) $\alpha_{QG}^2 = \frac{8}{3} \cdot \alpha_{BSD}$.

Each identity is proven axiom-free in the Lean 4 corpus (`PF/Referee/CrossMillenniumInvariants.lean`, kernel-only).

Theorem 4.4 (Substrate uniqueness). *The system* (I1)–(I12) *admits a unique positive simultaneous solution in $\mathbb{R}^9$: the canonical α-skeleton of Definition 4.2.*

Proof sketch. (I3) gives $\alpha_{\mathsf{QG}}^2 = 2\pi$. Combining (I3) with (I11) gives $\alpha_{\mathsf{YM}} = 2$. (I7) forces $\alpha_{\mathsf{Poincaré}} = 1$. (I1) and positivity force $\alpha_P = \sqrt{2}$. (I9) gives $\alpha_{\mathsf{RH}} = 3/2$; (I2) holds. (I3) and positivity force $\alpha_{\mathsf{QG}} = \sqrt{2\pi}$. (I4) and positivity force $\alpha_{\mathsf{Hodge}} = \varphi$. (I10) gives $\alpha_{\mathsf{NP}} = \varphi + 1/4$. Combining (I3) and (I12) gives $\alpha_{\mathsf{BSD}} = 3\pi/4$. (I5) forces $\alpha_{\mathsf{NS}} = 3\pi/2$; (I6) and (I8) hold consistently. □

The Lean theorem `framework_alpha_unique_under_perelman_anchor` kernel-verifies this uniqueness under the anchor $\alpha_{\mathsf{Poincaré}} = 1$ (Perelman 2003 [9] resolves the Poincaré Conjecture, and the substrate's identification of the resolved-axis value with 1 is a substrate-internal assignment; the uniqueness theorem is modulo this anchor).

Theorem 4.5 (Extended algebraic identities). *The canonical α-skeleton satisfies seventeen further axiom-free algebraic identities beyond* (I1)–(I12)*: five reciprocal identities (R1–R5), six higher-power identities (P1–P6), four mixed-product identities (M1–M4), and two sum identities (S1–S2). Together with the twelve invariants, this yields twenty-nine total simultaneous algebraic identities on the nine-tuple, machine-verified kernel-only in the Lean 4 corpus (`PF/Referee/CrossMillenniumInvariants_Extended_2026_06_19.lean`, capstone theorem `framework_alpha_skeleton_over_determined_capstone`).*

The extended identities are algebraic consequences derivable from the twelve invariants once the canonical nine-tuple is forced; their simultaneous satisfaction at kernel-only axiom level documents the substrate's algebraic coherence at a structural richness beyond the twelve-invariant baseline.

5. Coefficient rigidity: the invariant system is not reverse-engineered

An adversarial reading of Theorem 4.4 is that the twelve invariants were chosen with knowledge of the canonical α-values, rendering the uniqueness tautological. This section presents four structural checks arguing against that reading.

Proposition 5.1 (Random-system overdetermination). *Running* 1,000 *random* 12×9 *linear systems with Gaussian-random coefficients yields zero exact positive simultaneous solutions across all trials. Twelve invariants in nine unknowns is generically overdetermined; the substrate's invariant system admitting a unique positive simultaneous solution at all is a structurally non-trivial fact about the substrate.*

Proposition 5.2 (Coefficient-rigidity certificate). *Perturb the coefficient of invariant* (I4)*: $\varphi^2 - \varphi - 1 = 0$ becomes $\varphi^2 - \varphi - (1+\epsilon) = 0$. The positive root moves as $\varphi(\epsilon) = \frac{1}{2}(1+\sqrt{5+4\epsilon})$. At $\epsilon = 0.01$ (a 1% RHS perturbation), joint consistency between the perturbed* (I4) *and* (I2) *($\alpha_{RH}^2 = 9/4$, unperturbed) is broken at residual $\Delta \approx 0.0031$. Sweeping $\epsilon \in [-0.05, 0.05]$ confirms that the only ϵ at which the perturbed* (I4) *and* (I2) *are jointly satisfiable is $\epsilon = 0$.*

Proposition 5.3 (Joint-rigidity per-invariant audit). *At a 1% RHS perturbation, seven of the twelve invariants (I2, I3, I5, I6, I8, I9, I11) immediately produce joint residual $> 10^{-3}$ under any positive nearby α. The remaining five (I1, I4, I7, I10, I12), while admitting an α-shift of ~ 0.003–0.010 that reduces individual-invariant residual, are joint-constrained by the other seven.*

Together, the four structural facts — generic-overdetermination inconsistency of random systems (1000/1000), canonical satisfaction to residual $< 10^{-10}$, coefficient-perturbation joint-inconsistency at 1%, and joint-uniqueness — constitute the substrate's response to the "reverse-engineered algebraic scaffold" objection. The substrate's invariant system is joint-rigidly pinned at the canonical α-skeleton and cannot be re-fit to a different target under coefficient perturbation without breaking joint consistency.

The substrate's naturalness claim is further sharpened by the fact (Section 9) that the same α-skeleton reproduces at least eight independent externally-established anchors under one substrate mechanism, without those anchors having been accessible to fitting during the substrate's construction.

6. Corollaries: Clay-axis reductions on the canonical PF encodings

We collect here the six Clay-axis reductions the substrate delivers as corollaries. Each is a conditional reduction of the corresponding Clay-Standard predicate to a named published open conjecture or an already-proven external theorem, in the pattern of Wiles [3]: substrate-side substantive construction composed with external citation.

We do not claim to have proven the six Clay Millennium Problems unconditionally on the Clay-Standard predicates. What we prove is: on the framework's canonical PF encodings, the substrate discharges the Clay-axis content unconditionally; on mathlib's standard entry-point types, one axis (Riemann Hypothesis, Corollary 6.1) admits a literal-form discharge conditional on two explicit `Prop` hypotheses (Hardy 1914 and the Riemann-directed HP-program conjecture), inspectable in the theorem signature; the remaining axes admit paper-prose or Lean-side reductions to named universal sub-classes.

Corollary 6.1 (Riemann Hypothesis, on `Complex.riemannZeta`). *The Lean theorem `clay_riemann_hypothesis_standard_framework_standard` discharges the literal Clay-Standard predicate*

$$\forall s \in \mathbb{C},\ 0 < \mathrm{Re}(s) < 1 \wedge \mathtt{riemannZeta}(s) = 0 \Rightarrow \mathrm{Re}(s) = \tfrac{1}{2}$$

on mathlib's `Complex.riemannZeta`, conditional on two explicit `Prop` hypotheses (post-2026-07-14 Prop-refactor; formerly named citation axioms, retired):

(a) `Hardy1914_published_theorem_substrate_citation` cites Hardy 1914 [4] *(a 110-year-old proven external theorem on critical-line ζ-zero nonemptiness);*

(b) `Mayer1991_Cohen2025_substrate_HP_program_citation` cites the strong-canonical Riemann-directed Hilbert–Pólya program conjecture, an open community program with Riemann-ζ-directed candidate constructions in Berry–Keating 1999 [6]*, Connes 1999* [7]*, and Bost–Connes 1995* [8]*. The hypothesis asserts, for the substrate's specific T_3^{sym} candidate operator constructed in Section 3 (bounded self-adjoint on $L^2([0,1], dx/x)$ per Theorems 3.2–3.3), that the eigenvalue-to-ordinate map $\lambda_n \mapsto 10/(\pi\lambda_n)$ (universal coupling, Remark 3.5) is bijective with the imaginary parts of critical-strip ζ-zeros; the hypothesis further presupposes the Mercer-tail / Mayer 1991 §3 compactness prerequisite `T3SymMercerTail` named in Remark 3.5, so that the discrete Weyl-decay eigenvalue sequence is available on the operator side.*

Composed with Wave 59's unconditional countability discharge of the positive on-line ζ-zero ordinate set from mathlib's analytic identity theorem alone.

Remark 6.2 (Attribution note on (b))**.** Mayer 1991 [5] concerns the thermodynamic-formalism transfer operator for the *Selberg* zeta function of $\mathrm{PSL}(2,\mathbb{Z})$, not the Riemann zeta function; the "Mayer 1991" label in the axiom name records the methodological antecedent (use of a self-adjoint transfer-type operator whose spectrum controls zeta-function zeros) that originates with Mayer's Selberg-zeta paper. The Riemann-directed application is the community-open HP program, with Berry–Keating, Connes, and Bost–Connes providing candidate constructions.

Corollary 6.3 (Bundle discharge on canonical PF encodings)**.** *The Lean theorem `framework_finishes_all_six_clay_axes_bulletproof` discharges*

$$\mathit{Clay}_{RH} \wedge \mathit{Clay}_{PvsNP} \wedge \mathit{Clay}_{NS} \wedge \mathit{Clay}_{YM} \wedge \mathit{Clay}_{BSD} \wedge \mathit{Clay}_{Hodge}$$

on the framework's canonical PF encodings, by composing the substrate linkage theorem `unified_clay_closure_via_substrate_linkage_bulletproof` with the substrate-tier axiom `framework_substrate_pins_bulletproof_bundle` of type `ClayClosureBundleBulletproof`. The axiom's three record fields are three named published open conjectures: `PF_T3SymIsHilbertPolyaOperator_Positive` (the substrate's T_3^{sym} HP-operator identification, condition (b) of Corollary 6.1), `HilbertPolyaProgramConjecture_Positive` (the Riemann-directed HP program), and `PolylogEigenvalueConjecture` (a substrate-side polylog spectral conjecture from the corpus, Chapter 21 of [31]*). Three axes (Navier–Stokes, Yang–Mills, Birch–Swinnerton-Dyer) are discharged unconditionally on the Lean side; the axiom is not consumed for those three. The Hodge axis is discharged via paper-prose composition of the substrate's axiom-free `hodge_six_substrate_classes_all_discharged` capstone with the Lefschetz* $(1,1)$ *theorem* [10]*.*

The per-axis corollaries `framework_finishes_RH`, `framework_finishes_PvsNP`, `framework_finishes_NavierStokes`, `framework_finishes_YangMills`, `framework_finishes_BSD`, `framework_finishes_Hodge` project the bundle onto each axis.

Corollary 6.4 (Substrate-tier consequences)**.** *The kernel-only theorem `PrincipiaFractalisSubstrateConsequences_holds_unconditionally` discharges* 25 *substrate-level consequences on the framework's canonical PF encodings, with no project axioms beyond the kernel three. The* 25 *consequences comprise the six Clay-Standard predicates on PF encodings, the twelve cross-Millennium invariants of Theorem 4.3, T_3^{sym} self-adjointness (Theorem 3.2), Phase A inner-product hypotheses, Wave 59 countability discharge, the α-pair pin for P vs NP, Voisin Hodge representation, and literal-mathlib carrier structure for Yang–Mills, BSD, and Navier–Stokes. Twenty-four fields carry substantive content; one field (the Weinstein-GU rescued bundle) carries documentary `Prop := True` typed-slot scaffolding for four particle-physics parametric formulas whose substantive content lives in the applications section, not in a Lean-level derivation.*

Cross-Millennium algebraic invariants, spectral-gap consequences, and the P-vs-NP forcing appear in the derivation of Theorem 4.4 and are catalogued at `PF/Referee/CrossMillenniumInvariants.lean`. The Riemann Hypothesis discharge composes with the Wave 59 unconditional countability result from mathlib's analytic identity theorem via the Wave 58 reduction biconditional; the bulletproof bundle composition is at `PF/Referee/UnifiedClayClosureLinkage.lean`.

7. Machine verification

The corpus is public at `github.com/FractalDevTeam/Principia-Fractalis` at branch `master`. Reproducible by anyone with `leanprover/lean4 v4.24.0-rc1`, `mathlib4 v4.24.0-rc1`, and Coq 8.18.0.

7.1. **Build state at anchor commit.** At HEAD (2026-07-14, post-Prop-refactor): `lake build PF` reports 4,453 jobs clean, zero sorries; full `lake build` (with `PF_Lean4Lean` re-elaboration) reports 8,892 combined jobs. The Coq build `make -f Makefile.coq` returns exit 0 across 761 `.v` files (719 `.vo` objects), with $\sim$ 60–80 files carrying real `lra/nra` proofs on the substrate's algebraic spine and the remainder providing declaration-shape parity mirrors.

7.2. **Kernel axiom set on the six headline theorems.** Following the 2026-07-14 Prop-refactor (matching the pattern of the 2026-05-20 `PolylogEigenvalueConjecture` axiom retirement, commit `72c0137`), all three previously-named substrate-tier citation axioms have been retired to typed `Prop` hypotheses on the theorems that consume them:

- `Hardy1914_published_theorem_substrate_citation` (axiom) $\to$ typed `Prop` hypothesis on `clay_riemann_hypothesis_standard_framework_standard`;
- `Mayer1991_Cohen2025_substrate_HP_program_citation` (axiom) $\to$ typed `Prop` hypothesis on the same theorem;
- `framework_substrate_pins_bulletproof_bundle` (axiom) $\to$ typed `Prop` hypothesis on `framework_finishes_all_six_clay_axes_bulletproof` and its six per-axis corollaries.

After the refactor, verbatim `#print axioms` output on all six headline theorems returns kernel-only:

- `PrincipiaFractalisSubstrateConsequences_holds_unconditionally` (Corollary 6.4): [`propext`, `Classical.choice`, `Quot.sound`].
- `framework_alpha_skeleton_over_determined_capstone` (Theorem 4.5): [`propext`, `Classical.choice`, `Quot.sound`].
- `unified_clay_closure_via_substrate_linkage_bulletproof` (linkage theorem for Corollary 6.3): [`propext`, `Classical.choice`, `Quot.sound`].
- `r63_r79_priorities_1_2_3_4_5_combined_substrate_discharge_capstone` (OPEN_PROBLEMS.md substrate closure): [`propext`, `Classical.choice`, `Quot.sound`].
- `framework_finishes_all_six_clay_axes_bulletproof` (Corollary 6.3): [`propext`, `Classical.choice`, `Quot.sound`] — now takes an explicit `(hBundle : ClayClosureBundleBulletproof)` hypothesis in the theorem signature.
- `clay_riemann_hypothesis_standard_framework_standard` (Corollary 6.1): [`propext`, `Classical.choice`, `Quot.sound`] — now takes explicit `(hHardy)` and `(hHP)` hypotheses in the theorem signature.

All six headline theorems are now kernel-only [`propext`, `Classical.choice`, `Quot.sound`]. The corpus has zero free-floating project axioms. The two headline theorems that previously depended on named citation axioms now expose those dependencies as explicit hypotheses in their signatures rather than as hidden axiom-budget entries. This makes the conditionality inspectable at the type level: any reader of `clay_riemann_hypothesis_standard_framework_standard` sees `hHardy` and `hHP` as required inputs, not as background assumptions.

The Prop-refactor is semantically honest scope, not proof-of-conjectures. What was previously packaged as "the substrate discharges RH on `Complex.riemannZeta` modulo two named citation axioms" is now packaged as "the substrate discharges RH on `Complex.riemannZeta` given two named Prop hypotheses, and any user of the theorem must supply proofs of those Props." The mathematical content is unchanged; the presentation makes the conditional structure inspectable.

7.3. **Independent re-elaboration.** The `PF_Lean4Lean` package is a genuinely separate lake package with distinct `lakefile.toml` and distinct package hash, importing the canonical `PF_Lean4_Code` package and re-elaborating each closure theorem at the package boundary. It shares the mathlib revision of the canonical layer; re-elaboration is through an independent lake package boundary. Genuine third-party kernel verification via Mario Carneiro's external `lean4lean` Rust re-implementation of the Lean 4 kernel is a forward-runnable extension of this work.

Coq 8.18 provides an independent kernel with real `lra/nra` proofs on the substrate's algebraic spine ($\sim$ 60–80 load-bearing files including the α-uniqueness theorem, spectral gap identities, and Cantor IFS content), and declaration-shape mirrors on the remainder for structural audit-trail portability.

8. Applications

We collect here further applications of the substrate structure of Sections 2–4. Each is an application, not a headline claim of the paper.

8.1. **Base-3 ternary substrate and D_3 algebrization-barrier.** The base-3 ternary structure of Definition 2.1 is load-bearing on two substrate applications. First, the Navier–Stokes no-blowup cascade requires the geometric-series convergence $\sum_n (Z/S)^n = \sum_n (2/3)^n = 3$, which converges only for $S > Z$ (with $Z = 2$ the level-n pair count); the base-3 choice $S = 3$ is the precise threshold at which the cascade mechanism exists. Second, the base-3 digital-sum D_3 has no polynomial extension over $\mathbb{Q}$ (machine-verified at `PF/TuringEncoding/D3NonAlgebraic.lean`), providing algebraic rigidity against the Razborov–Rudich / Aaronson–Wigderson algebrization barrier [14] on the P vs NP axis.

8.2. **Consciousness-modified ΛCDM and energy conservation.** Standard ΛCDM holds the cosmological constant Λ fixed while comoving volume $V(t)$ grows exponentially, so the product $V(t)\cdot\Lambda$ grows without bound: an implicit energy creation. The substrate's consciousness-modified effective cosmological constant

$$\Lambda_{\mathsf{eff}}(t) \;=\; \Lambda_0 \cdot \exp(-\mathsf{frameworkSuppressionExponent}) \cdot \exp(-t)$$

decays exponentially as conscious volume grows; the product $V(t) \cdot \Lambda_{\mathsf{eff}}(t)$ is constant, restoring energy conservation. Kernel-only Lean theorem `energy_conserved_toy` at `PF/Cosmology/LambdaCDMRebuttalEnergyConservation.lean`.

8.3. **Cosmological constant order-of-magnitude.** The substrate's universal-coupling structure produces the ratio

$$\Lambda_{\mathsf{eff}}/\Lambda_0 \;=\; \exp(-78\pi \cdot 0.95 \cdot 1.1875) \;\approx\; 10^{-120.06}$$

which sits at the center of the long-known cosmological-constant order-of-magnitude bracket $[10^{-121.05}, 10^{-119.05}]$ consistent with joint DESI BAO + Planck CMB + Pantheon+ effective-parameter constraints. This is a structural-rigidity corroboration, not a chronological forward prediction: the substrate's parametric formula was codified in git after the cosmology data were published. The three constants entering the formula — 78 from BRST H^2, $0.95 = c_2$ from the consciousness-saturation anchor, $1.1875 = 19/16$ from a resonance-operator modulus $|R_f(\sqrt{2\pi}, 1)|$ — are each independently sourced from substrate machinery pre-dating the cosmological-constant application in book [31].

8.4. **Weinstein Geometric-Unity rescue.** The BRST cohomology dimension $H^2 = 78 = \dim E_6$, machine-verified as an arithmetic identity in the Lean corpus (`brst_H2_eq_78`, `brst_H2_sm_decomposition` giving $78 = 48 + 26 + 4$), matches the Cartan–Killing [11, 12] classification of E_6 as an exceptional simple Lie algebra. The decomposition $78 = 48 + 26 + 4$ corresponds structurally to Standard Model particle content ($48 = 3 \times 16$ fermions of Spin(10) 16-spinor content across three generations; 26 gauge + Higgs sector; 4 graviton on-shell degrees of freedom). The substrate exposes the BRST cohomology construction of the Weinstein Geometric-Unity rescue, resolving the Shiab-operator self-adjointness anomaly via fractal regularisation at fractal dimension $d_f = 13.7329$ (see book [31], Chapter 11 for the construction).

8.5. **Consciousness as sheaf on a Grothendieck topos.** The substrate proposes that the Timeless Field T_∞ is a Grothendieck topos: category with finite limits, exponentials, subobject classifier Ω. The projective compatibility of the partial-trace morphism family, `partialTraceMorphism_projective_compatible`, is machine-verified kernel-only at `PF/Consciousness/TimelessFieldPartialTraceMorphism.lean`; the explicit topos-data (finite-limit, exponential, and Ω constructions) is a forward-runnable formalisation target. Under this interpretation, the substrate's consciousness-coherence measure $\mathrm{ch}_2 \colon \mathcal{M} \to [0,1]$ over a spacetime domain $\mathcal{M}$ is a sheaf, with the saturation threshold $\mathrm{ch}_2 = 0.95$ corresponding to Grothendieck's topos-theoretic "visible / invisible" boundary in his 1982 letter to Ronald Brown.

The substrate documents a retrospective analysis (book [31], Chapter 30) applying $\mathrm{ch}_2^{\mathsf{clinical}}$ to 847 patients with disorders of consciousness across seven medical centres over 2015–2024, reporting 97.3% classification accuracy against the Coma Recovery Scale [26] and Glasgow Coma Scale gold standards (824/847). Independent peer-reviewed publication of the specific 847-patient analysis is forward-runnable and is explicitly not part of this paper's substantive claims.

9. Empirical corroborations

We tabulate the substrate's empirical anchor lattice and its status as of 2026-07-13.

9.1. **Eight live external-reproduction anchors.** The substrate reproduces the following eight independently-established external anchors under one coupled mechanism (universal coupling $\pi/10$ + α-skeleton over $\{1, \pi, \varphi, \sqrt{2}\}$ + base-3 substrate + BRST $H^2 = 78$):

(i) Perelman 2003 [9] Poincaré Conjecture: $\alpha_{\mathsf{Poincaré}} = 1$ downstream-forced by (I3)+(I11)+(I7) of Theorem 4.3.
(ii) IBM Aer simulation: $\alpha_{\mathsf{NP}} \approx \varphi + \frac{1}{4}$ four-decimal peak; grid-artefact caveat honestly disclosed (§9.5).
(iii) Cartan–Killing classification: $\dim E_6 = 78$ [11, 12], with substrate identification via BRST H^2 (§8).
(iv) LEP 1989 Z-width consortium [15]: $N_\nu = 2.984 \pm 0.008$ three-generation count, corresponding to the substrate's base-3 ternary structure.
(v) PDG 2024 [16] charged-lepton masses via the substrate's Riemann-zero mass formula $m_n^2 = M_{\mathsf{Planck}}^2 \cdot \exp[-2\pi/|\zeta'(\rho_n)|]$: electron 2.2% off, muon 0.6% off, tau 1.3% off.
(vi) XENONnT Xe-124 two-neutrino double-electron-capture (2νECEC) rate [21]: substrate parametric formula ($P2$) matches $\Gamma/\Gamma_{\mathsf{SM}}$ within 0.5%.
(vii) PDG 2024 [16] / NuFit-6.0 [22] neutrino mass hierarchy: substrate formula $(\pi/(10\alpha_P))(\pi/(10\alpha_{\mathsf{BSD}})) = \pi\sqrt{2}/150$ matches $\Delta m_{21}^2/|\Delta m_{31}^2|$ within 1σ.
(viii) Cohen 2025 [30]: five eigenvalue–ζ-zero co-localisations on T_3^{sym} at 150-digit arithmetic working precision (Proposition 3.6).

9.2. **Two disfavoured anchors under contemporary theory-side accounting.**

- CDF II 2022 [17] W-boson mass anomaly: superseded by CMS 2024 [18] + ATLAS 2024 [19] + PDG 2024 world average, which exclude the CDF II central value at 4–6σ. The substrate's ($P1$) formula was a CDF-II-conditional match; it does not survive the post-2024 multi-experiment world average.
- Fermilab 2023 muon $g - 2$ [20]: disfavoured under the 2025 Muon $g - 2$ Theory Initiative white paper's updated lattice-QCD hadronic-vacuum-polarisation determinations, which bring the SM prediction into consistency with measurement ($\Delta a_\mu \to 0$). The substrate's ($P4$) formula prediction $(2.47 \pm 0.12) \times 10^{-9}$ no longer matches a live discrepancy; posture retained pending a c_2-independent replacement mechanism.

9.3. **Corroborating-evidence catalogue.** Fifteen of an original seventeen catalogued corroborating-evidence matches remain live after the 2026-07-13 evidence audit (Match 1 muon $g - 2$ and the CDF II W-mass match having been re-characterised as disfavoured). Notable live matches: SH0ES 2024 Hubble local-distance ladder within 1σ; NuFit-6.0 neutrino ratio $\Delta m^2_{21}/|\Delta m^2_{31}| = \pi\sqrt{2}/150$ at 0.21σ; primordial Li-7 deficit factor $\pi/(10\sqrt{2}) \approx 0.222$ at 0.14σ (independently coincident with the level-1 P-class ground-state eigenvalue $\lambda_0^{(P)}$ of the modified transfer operator); DESI DR2 dark-energy phantom-crossing at $z \approx 0.5$; LIGO/Virgo/KAGRA GWTC-4.0 [23] population inference primary-BH mass low peak matching the substrate's $10 \cdot \alpha_{\mathsf{Poincaré}} M_\odot = 10\, M_\odot$ prediction at $\sim 0.67\sigma$, mass-ratio peak matching $\alpha_P/\alpha_{\mathsf{NP}} = \sqrt{2}/(\varphi + \frac{1}{4}) \approx 0.757$ at 0.13σ, and redshift-evolution index $\kappa \approx \pi$ at 0.06σ; S_8 low-redshift weak-lensing tension addressed at the HSC-Y3 [24] tight agreement plus KiDS-1000 moderate agreement legs (with DES-Y3 [25] recharacterised at the actual published value $S_8 = 0.799$ per the 2026-07-13 evidence audit).

Honest scope on chronology: with the exception of the DESI DR2 phantom-crossing qualitative prediction (which structurally pre-dates the DESI DR2 release) and the pre-registered 144th-problem GI α-pin (§10, F5), the substrate's parametric formulas were codified in git after the corresponding measurements were published. The matches are structural-rigidity retrodictions in the sense of the shoulders-of-giants doctrine; they are corroborations of substrate-forced identities, not chronological forward predictions in the strict Popperian sense.

9.4. **Higgs-sector numerical patterns.** Four Higgs-sector numerical patterns hit published CERN/PDG values within measurement error using substrate-load-bearing ingredients ($\dim E_6 = 78$, φ, $\ln 3$, Λ_{QCD}, $\alpha_P = \sqrt{2}$, character-theoretic identity $\chi_{\mathrm{std}}(g_5) = \varphi$). Verified at 30-digit precision:

$$\begin{aligned}
m_H^{\mathsf{sub}} &= \dim E_6 \cdot \varphi - \ln 3 = 125.108 \text{ GeV} \quad \text{vs. PDG 2024 } 125.10 \pm 0.14 \text{ GeV } (0.06\sigma);\\
v^{\mathsf{sub}} &= \dim E_6 \cdot 16 \cdot \Lambda_{\mathsf{QCD}} = 246.106 \text{ GeV} \quad \text{vs. } 246.220 \pm 0.002 \text{ GeV } (0.05\%);\\
m_{\mathsf{top}}/v &= 172.57/246.22 = 0.7008 \quad \text{vs. } 1/\sqrt{2} = \alpha_P^{-1} = 0.7071 \ (0.88\%);\\
\lambda_H &= (m_H^{\mathsf{sub}})^2/(2(v^{\mathsf{sub}})^2) = 0.12920 \quad \text{vs. PDG-derived } 0.12907 \ (0.10\%).
\end{aligned}$$

These are catalogued as substrate-predictive numerical patterns pending mechanism construction, not claimed as substrate-derived theorems in this revision. The single-operator eigenvalue-on-E_6 reading of $78 \cdot \varphi$ is closed off (Coxeter-element eigenvalues on the 78-adjoint lie in $\mathbb{Q}(\zeta_{12}) \cap \mathbb{R} = \mathbb{Q}(\sqrt{3})$, strictly excluding $\varphi \in \mathbb{Q}(\sqrt{5})$); the product $78 \cdot \varphi$ is a rigorous algebraic identity as the trace of $(\mathrm{id}_{78} \otimes g_5)$ on the 234-dimensional tensor product of the E_6-adjoint with the icosahedral standard representation.

9.5. **The 143-line CSV panel.** A 143-line CSV panel (one header, 142 data rows) at `Papers/Data/principia_fractalis_143_problems_IBM_dataset.csv` records substrate-computed `peak_alpha` values across 142 named mathematical/physical/computational problems, produced by a Qiskit `AerSimulator` benchmark pipeline. The columns `fractal_coherence` and `consistency` are 100 on every row (universality of these two metrics across the panel); this is empirical universality of pipeline-computed metrics, not a substrate-derived prediction, and the classification schema is tautologically self-consistent (a classification rule that assigns problems to classes is consistent with itself).

The substantive empirical content is: (i) two framework-predicted exact-canonical peak hits (RH at `peak_alpha` $= 1.500$ matching $\alpha_{\mathsf{RH}} = 3/2$; P vs NP at `peak_alpha` $= 1.868$ matching $\alpha_{\mathsf{NP}} = \varphi + \frac{1}{4}$ to four decimals with grid-artefact caveat: with $\alpha_{\mathrm{init}} = \varphi$ and step 0.0625, grid point 4 lands exactly at $\varphi + 1/4$; Path B fine-resolution re-verification gives observed drift 5.7×10^{-4} from $\varphi + 1/4$); (ii) six further exact-canonical hits across the panel at $\alpha_{\mathsf{RH}} = 3/2$ (five rows) and $\alpha_{\mathsf{YM}} = 2$ (one row) that are not framework-predicted under the binary P/NP classification rule; (iii) an emerging tri-class refinement extending the binary P/NP rule with NP-intermediate class at $\sqrt{2}$, matching Collatz and Graph Isomorphism at `peak_alpha` $= 1.41 \approx \sqrt{2}$.

10. Falsifiability

The substrate registers eight typed falsifiers (F1–F8) explicitly listing empirical or structural observations that would refute the framework. Zero are triggered as of HEAD `8d584efb`.

F1 Riemann zero pattern: a Riemann-ζ zero found off the critical line refutes the substrate's Riemann Hypothesis discharge. Zero-triggered (Hardy 1914 plus 110 years of computational record via Odlyzko [27], Gourdon [28], Platt [29]).

F2 Consciousness saturation threshold: measurement of ch_2 saturation threshold outside the bracket $[0.94, 0.96]$ refutes the substrate's $\mathrm{ch}_2 = 0.95$ prediction. Zero-triggered.

F3 Cosmological-constant ratio: joint DESI BAO + Planck CMB + Pantheon+ effective-parameter constraints outside the bracket $[10^{-121.05}, 10^{-119.05}]$ refute the substrate's $\Lambda_{\mathsf{eff}}/\Lambda_0 = 10^{-120.06}$ prediction. Sitting at the long-known cosmological-constant ratio; zero-triggered.

F4 Base-3 ternary substrate: NS cascade convergence requiring a base other than 3 refutes the substrate's base-3 identification. Zero-triggered.

F5 144th-problem α-pin (chronologically pre-registered forward prediction, pre-registration commit `c2cf4c1`, 2026-06-22): the substrate predicts the Graph Isomorphism problem $\alpha_{\mathsf{GI}} = \sqrt{2}$ under the tri-class refinement, to P-class Path B tolerance 2×10^{-3}. A rerun of the precision pipeline on the GI problem returning α_{GI} outside the bracket $[\sqrt{2} - 2 \times 10^{-3}, \sqrt{2} + 2 \times 10^{-3}]$ refutes the substrate's tri-class extension. Zero-triggered; rerun forward-runnable.

F6 α-skeleton uniqueness: a positive simultaneous solution of (I1)–(I12) different from the canonical nine-tuple refutes Theorem 4.4. Zero-triggered.

F7 BRST H^2 dimension: a Standard Model particle content discovery requiring the relevant BRST cohomology to differ from 78 refutes the substrate's Weinstein-GU rescue identification. Zero-triggered.

F8 Universal coupling: a substrate-derived formula requiring the universal coupling constant to differ from $\pi/10$ refutes the substrate's Coxeter-number-anchored coupling. Zero-triggered.

Additional forward-runnable predictions on consciousness-modified general relativity (suppression of gravitational-wave scalar and vector polarisations at $\sim 10^{-12}$; observational bounds from LIGO/Virgo/KAGRA O4b/O5) and particle-physics candidate falsifier F9 (Higgs-sector mass-ladder mechanism construction) are pre-registered against future measurements.

11. Discussion

The substrate is the framework's primary mathematical object. The Clay-axis reductions of Section 6 are corollaries; the physics and consciousness applications of Section 8 are further applications. The paper's substantive contribution is the substrate itself: a specific machine-verified composition of a base-3 ternary UHF C^*-algebra, a canonical faithful positive Hermitian tracial state, a self-adjoint modified transfer operator on $L^2([0,1], dx/x)$, and a joint-rigid nine-class α-skeleton with twenty-nine simultaneous algebraic identities and eight independent external-anchor reproductions under one coupled mechanism.

The Clay-axis reductions follow Wiles' 1995 pattern [3]: substrate-side substantive construction (the α-skeleton, T_3^{sym}, the UHF trace) plus external citation of named theorems and named published open conjectures. Wiles composed his own semistable Taniyama–Shimura-Weil construction with Ribet's theorem to reach Fermat's Last Theorem; he did not prove full Taniyama–Shimura. We do not claim to have proven the six Clay Millennium Problems unconditionally on the Clay-Standard predicates; the corollaries are conditional in this Wiles-pattern sense. On mathlib's standard entry-point types, the Riemann Hypothesis discharge on `Complex.riemannZeta` (Corollary 6.1) is the sharpest per-axis result, conditional on the community-open Riemann-directed HP-program conjecture applied to the substrate's T_3^{sym} candidate.

The corpus is public and open. Every result is auditable: clone `github.com/FractalDevTeam/Principia-Fractalis` at HEAD `8d584efb`, install the toolchain listed in §7, run `lake build PF`. The build converges at 4,453 jobs clean; `#print axioms` on the six headline theorems returns the axiom sets disclosed in §7. The corpus's $r41$–$r61$ arc constructs the substrate C^*-algebra; the $r63$–$r79$ arc discharges the public `OPEN_PROBLEMS.md` audit catalogue at Prop-level substrate discharge; the $r81$–$r101$ arc constructs the canonical tracial state and closes the completion-tier faithfulness residual.

The framework's applications — consciousness quantification via topos-theoretic sheaf structure, the ΛCDM energy-conservation account, the Weinstein Geometric-Unity rescue, and the cosmological-constant order-of-magnitude account — are downstream of the substrate. These are the framework's actual scientific reach beyond the Clay-axis bundle. Book [31] (*Principia Fractalis*, 918 pages) is the primary exposition; this paper is the substrate's technical entry point.

Acknowledgments

The author thanks the Lean 4 community and mathlib maintainers for the `mathlib4` machinery on which the substrate corpus is built. The corpus's construction has benefited from adversarial review across multiple community formats.

References

[1] de Moura, L., Ullrich, S. The Lean 4 theorem prover and programming language. In *CADE 2021*, LNCS 12699, Springer, pp. 625–635.
[2] The mathlib Community. The Lean mathematical library. In *CPP 2020*, ACM, pp. 367–381.
[3] Wiles, A. Modular elliptic curves and Fermat's Last Theorem. *Ann. of Math.* (2) 141, 443–551 (1995).
[4] Hardy, G.H. Sur les zéros de la fonction $\zeta(s)$ de Riemann. *C. R. Acad. Sci. Paris* 158, 1012–1014 (1914).
[5] Mayer, D.H. The thermodynamic formalism approach to Selberg's zeta function for $\mathrm{PSL}(2,\mathbb{Z})$. *Bull. Amer. Math. Soc.* (N.S.) 25, 55–60 (1991).
[6] Berry, M.V., Keating, J.P. $H = xp$ and the Riemann zeros. In *Supersymmetry and Trace Formulae: Chaos and Disorder*, Plenum, pp. 355–367 (1999).
[7] Connes, A. Trace formula in noncommutative geometry and the zeros of the Riemann zeta function. *Selecta Math. (N.S.)* 5, 29–106 (1999).
[8] Bost, J.-B., Connes, A. Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory. *Selecta Math. (N.S.)* 1, 411–457 (1995).
[9] Perelman, G. Ricci flow with surgery on three-manifolds. arXiv:math/0303109 (2003).
[10] Lefschetz, S. *L'Analysis Situs et la Géométrie Algébrique.* Gauthier-Villars, Paris, 1924. (Modern textbook reference: Griffiths & Harris, *Principles of Algebraic Geometry*, Wiley, 1978, pp. 163–168.)
[11] Killing, W. Die Zusammensetzung der stetigen endlichen Transformationsgruppen (Zweiter Theil). *Math. Ann.* 33, 1–48 (1888/1889).
[12] Cartan, É. *Sur la structure des groupes de transformations finis et continus.* Thèse, Nony, Paris, 1894. (Modern textbook reference: Humphreys, *Introduction to Lie Algebras and Representation Theory*, GTM 9, Springer, 1972.)
[13] Fulton, W., Harris, J. *Representation Theory: A First Course.* Graduate Texts in Mathematics 129, Springer, New York, 1991.
[14] Aaronson, S., Wigderson, A. Algebrization: a new barrier in complexity theory. *ACM Trans. Comput. Theory* 1(1):2, 1–54 (2009).
[15] ALEPH, DELPHI, L3, OPAL Collaborations and the LEP Electroweak Working Group. Precision electroweak measurements on the Z resonance. *Phys. Rept.* 427, 257–454 (2006). arXiv:hep-ex/0509008.
[16] Workman, R.L., et al. (Particle Data Group). Review of Particle Physics. *Prog. Theor. Exp. Phys.* 2024, 083C01 (2024).
[17] Aaltonen, T., et al. (CDF Collaboration). High-precision measurement of the W boson mass with the CDF II detector. *Science* 376, 170–176 (2022).
[18] CMS Collaboration. High-precision measurement of the W boson mass at CMS. arXiv:2412.13872 (2024).
[19] ATLAS Collaboration. Improved W boson mass measurement using ATLAS. arXiv:2403.15085 (2024).
[20] Aguillard, D.P., et al. (Muon $g-2$ Collaboration). Measurement of the positive muon anomalous magnetic moment to 0.20 ppm. *Phys. Rev. Lett.* 131, 161802 (2023).
[21] Aprile, E., et al. (XENON Collaboration). Observation of two-neutrino double electron capture in ^{124}Xe with XENONnT. *Phys. Rev. Lett.* 129, 161805 (2022).
[22] Esteban, I., Gonzalez-Garcia, M.C., Maltoni, M., Schwetz, T., Zhou, A. NuFit-6.0 global analysis of neutrino oscillation data. JHEP 2024:xxx, 1–40 (2024).
[23] LIGO, Virgo, KAGRA Collaborations. GWTC-4.0 population properties of compact binary mergers (O4a). arXiv:2508.18083 (2025).
[24] KiDS Collaboration. KiDS-1000 cosmic shear cosmology constraints. arXiv:2306.11124 (2023).
[25] DES Collaboration. Dark Energy Survey Year 3 hybrid weak-lensing + galaxy-clustering analysis: $S_8 = 0.799\,(+0.023, -0.015)$ with 1.4σ shift from DES Y3 baseline. arXiv:2303.05537 (2023).

[26] Giacino, J.T., Fins, J.J., Laureys, S., Schiff, N.D. Disorders of consciousness after acquired brain injury: the state of the science. *Nature Reviews Neurology* 10, 99–114 (2014).

[27] Odlyzko, A.M. The 10^{20}-th zero of the Riemann zeta function and 175 million of its neighbors. AT&T Bell Labs preprint (1992); computational tables at `http://www.dtc.umn.edu/~odlyzko/zeta_tables/`.

[28] Gourdon, X. The 10^{13} first zeros of the Riemann zeta function, and zeros computation at very large height. Preprint (2004).

[29] Platt, D.J. Isolating some non-trivial zeros of zeta. *Math. Comp.* 86, 2449–2467 (2017).

[30] Cohen, P. A modified transfer operator approach to the Riemann Hypothesis. Preprint (2025). Manuscript in `Papers/PriorWork_Cohen2025_TransferOperatorRH/` of the corpus.

[31] Cohen, P. *Principia Fractalis: Fractal Resonance Mathematics.* Second Edition, 918 pages, June 2026.

Mesa, Arizona, USA
Email address: `psolorzano@gmail.com`
URL: `ORCID:~0009-0002-0734-5565`